No.

Date

Trịnh Gia Hiệp - 2352343 - W07

1. Factorization:

$$P(S, P, R, W, L) = P(S) \cdot P(P|S) \cdot P(W|P,R) \cdot P(L|W)$$

2. Is $P \perp R$: Yes

~~Card~~ There are 2 paths between P and R:

. $P \leftarrow S \rightarrow R$ is a fork at S. S is observed, so the path is blocked.

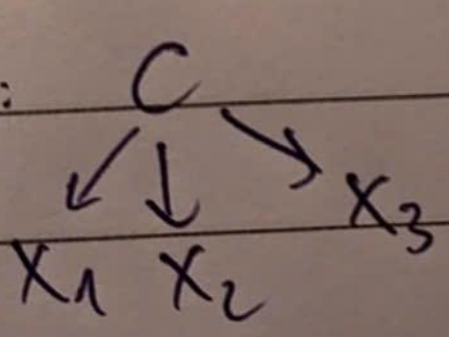
3. Is $P \perp R \mid S, L$? No

. $P \leftarrow S \rightarrow R$ is still blocked, S is observed.

. $P \rightarrow W \leftarrow R$ is ~~cccl~~ a collider at W, L is a descendant of W and is now observed → The collider is activated and the path is opened.

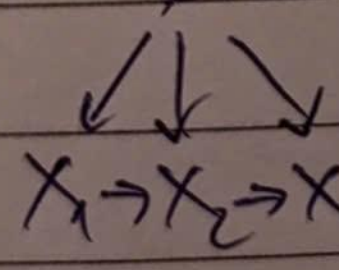
4. Naive Bayes and TAN.

- Naive Bayes:



$$P(C, X_1, X_2, X_3) = P(C) \cdot P(X_1|C) \cdot P(X_2|C) \cdot P(X_3|C)$$

- TAN:



$$P(C, X_1, X_2, X_3) = P(C) \cdot P(X_1|C) \cdot P(X_2|C, X_1) \cdot P(X_3|C, X_2)$$

Tan relaxes the ~~Naive~~ Naive Bayes assumptions that features are conditionally independent given the class.

HAI TIEN